DeepSS2GO

3
CrossSpecies/
CAFA3/
PredictNew/ fasta GO

output/ CS
pub_data/
redundancy/ data preprocessing *.npt* & .csv

Cross species

6 ARATH, ECOLI, HUMAN, MOUSE, MYCTU, YEAST, & ALL00

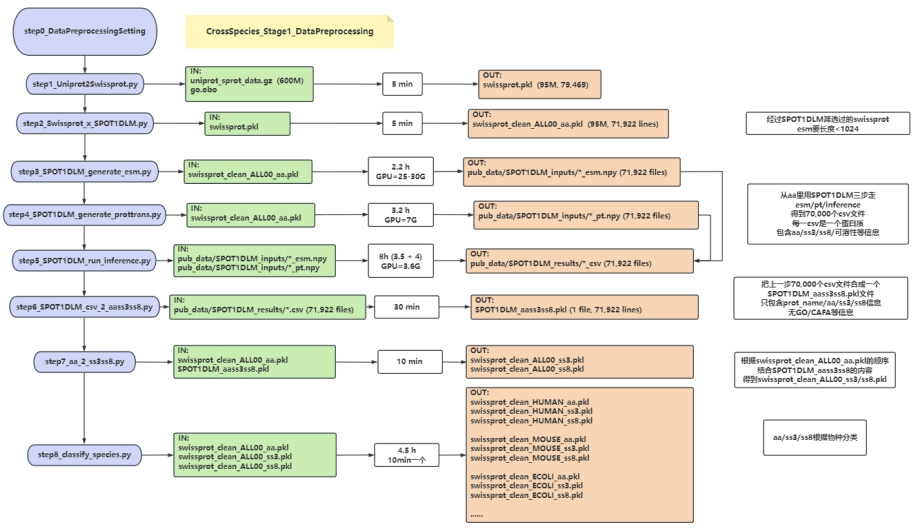

Figure 1: CrossSpecies_s1

s1_DataPreprocessing_CrossSpecies/:

- step1_Uniprot2Swissprot.py:
- step2_Swissprot_x_SPOT1DLM.py: 1024
- step3_SPOT1DLM_generage_esm.py: *esm.npy
- step4_SPOT1DLM_generate_prottrans.py: *pt.npy
- step5_SPOT1DLM_run_inference.py *.csv
- step6_SPOT1DLM_csv_2_aass3ss8.py: SPOT1DLM_aass3ss8.pkl
- step7_aa_2_ss3ss8.py: swissprot_clean_ALL00_ss3.pkl & swissprot_clean_ALL00_ss8.pkl
- step8_ClassifySpecies.py: ss3/ss8.pkl

CrossSpecies_s2

s2_TrainTest/:

& CrossSpecies - step1_SplitTrainTest_Terms: train & test terms - step2_Train: model + training.csv - step3_Test: predictions.pkl - step4_pkl2fa: - step5_Diamond4CrossSpecies: diamond_aa.res - step6_FindAlpha: Alpha - step7_EvaluateAlpha: Alpha Fmax, AUPR, Smin - step8_PredictAlpha: test_data.pkl results_bp/cc/mf.csv

& - step9_cpData_Diamond4New: Diamond - step10_Predict_New: aa/ss8

s2_TrainTest

CrossSpecies/s2_TrainTest/ global

pub_data TrainECOLI_TestYEAST_aa 1. CrossSpecies/ s2_TrainTest/ s2_TrainTest_TrainECOLI_TestYEAST_aa/ (global)

2. step0_TrainTestSetting_global.py

- 'device_ids': [0, 1]
- 'dir0': 'test_TrainHUMAN_TestHUMAN_aa_test/'
- 'aa_ss': 'aa'
- 'train_data': 'HUMAN'
- 'test_data': 'HUMAN'
- 'kernels': [8, 16, 24, ...]
- 'filters': [16, 32, 64, ...]

3. run_KernelX_FilterY.py output/ test_TrainECOLI_TestYEAST_aa/ (local = dir0) kernel&filter local e.g. DeepSS2GO_Kernel104_Filter8192_Ontsall/ (local = dir1) global cp local step0_TrainTestSetting_local.py i.e. cp -rf CrossSpecies/s2_TrainTest_TrainX_TestY_aa/ output/dir0/dir1/ e.g. cp -rf CrossSpecies/s2_TrainTest_TrainECOLI_TestYEAST_aa/ output/test_TrainECOLI_TestYEAST_aa/DeepSS2GO_Kernel104_Filter8192_Ontsall/

4. local step0_TrainTestSetting_local.py : step1-8 step1-3 step4-8 find_alpha

CrossSpecies_s3 ### s3_AlphaBeta/ alpha + beta + Diamond (s2) > & CrossSpecies - step1_cp_data.sh: aa&ss8 pkl/fa

cp - step2_Diamond4CrossSpecies.sh - step3_FindAlphaBeta.sh - step4_EvaluateAlphaBeta.sh: - step5_PredictAlphaBeta.sh results_bp/cc/mf.csv

& - step6_cpData_Diamond4New.sh - step7_PredictAlphaBeta_New.sh

s3_AlphaBeta :

1. s2_TrainTest_TrainALL00_TestALL00_aa/ss8 bp/cc/mf
2. bp aa K16F65536 ss8 K48F8192 s3_AlphaBeta/ cp s3_AlphaBeta_TrainALL00_TestALL00_bp_aaK16F65536_ss8K48F8192/
3. step1_cp_data.sh
   - path_aa="${path_base}output/best/test_TrainALL00_TestALL00_aa_DeepSS2GO_Kernel16_
   - path_ss8="${path_base}output/best/test_TrainALL00_TestALL00_ss8_DeepSS2GO_Kernel48
4. step1-5
   step3_FindAlphaBeta
   - ont(bp/cc/mf)
     -o bp –alpha-range '0, 101, 10' –beta-range '0, 101, 10'
   - e.g. bp a=0.2, b=0.3 range 0.1-0.3, 0.2-0.4 2
     -o bp –alpha-range '10, 31, 2' –beta-range '20, 41, 2' ((0.16, 0.28, -0.535361639858736))

CAFA3

TrainALL & Predict New

case 0 & case 1 " "

case 0: aa/ss8_model + Diamond

PredictNew/s1_DataPreprocessing_PredictNew/

1. fasta aa pub_data/data_new/new_aa.fa
2. s1_DataPreprocessing_New step1-8:

- step1_fa2pkl.py: new_aa.fa pkl
- step2_New_x_SPOT1DLM.py: aa >1024
- step3_SPOT1DLM_generate_esm.py:
- step4_SPOT1DLM_generate_prottrans.py:
- step5_SPOT1DLM_run_inference.py:

- step6_SPOT1DLM_csv_2_aass3ss8.py: new_SPOT1DLM_aass3ss8.pkl
- step7_aa_2_ss3ss8.py: new_clean_aa.pkl & new_clean_ss8.pkl
- step8_pkl2fa.py: fa

pub_data/data_new/
new_clean_aa.pkl new_clean_aa.fa
new_clean_ss8.pkl new_clean_ss8.fa

CrossSpecies/s2_TrainTest/step9-10

3. CrossSpecies/s2_TrainTest/step9_cpData_Diamond4New.sh 4 pkl/fa cp diamond e.g. output/test_TrainALL00_TestALL00_aa_DeepSS2GO_Kernel8_Filter65536_Ontsall/data/
4. CrossSpecies/s2_TrainTest/step10_Predict_New: aa/ss8

case 1: aa_model + ss8_model + Diamond

PredictNew/s1_DataPreprocessing_PredictNew/

case0 (PredictNew/s1_DataPreprocessing_PredictNew/)

CrossSpecies/s3_AlphaBeta/step6-7

3. CrossSpecies/s3_AlphaBeta/step6_cpData_Diamond4New.sh
4. CrossSpecies/s3_AlphaBeta/step7_PredictAlphaBeta_new.sh

And they all lived happily ever after!